

ARAS SELAHIYE

Photonics and Data Engineer

Dual MSc — Optics & Photonics + Computer Science · Machine learning for imaging · Health & defence R&D

+33 (0)7 82 39 24 76 | aras.selahiye@hotmail.com | [linkedin/aras-selahiye](https://linkedin.com/in/aras-selahiye) | github.com/Alltairas

Strasbourg, France | Turkish citizen | Military service: completed

Turkish (native) · English C1 (TOEIC 910/990) · French C1 (French Baccalauréat) · Spanish A2 (DELE)



Physicist working where **optical systems** meet **machine learning**. Two research placements in 2025–2026 building polarimetric imaging for the **diagnosis of conditions associated with preterm birth**: Mueller-matrix reconstruction pipelines on clinical data, and deep-learning models trained on medical images. Earlier, **electro-optic test and calibration at a Turkish defence electro-optics manufacturer** boresight alignment on thermal imaging systems. Works on full detector build-up, LabVIEW acquisition, large-scale image processing in Python and MATLAB.

PROFESSIONAL EXPERIENCE

Polarimetric Imaging Reconstruction & Data Pipeline — Research Intern

Apr – Sep 2026

ICube TRIO, UMR 7357 CNRS

Strasbourg, France

- Refactored a legacy MATLAB package into a **Mueller polarimetric imaging** reconstruction pipeline for biomarker extraction, with a Python metadata layer, on a **clinical dataset of 97 patients**.
- Designed a **statistical validation framework** quantifying inter-calibration drift.
- Built a **semantic segmentation** model for polarimetric images, distinguishing cervix epithelial types.

Polarimetric Imaging & Deep Learning, at POLADERME startup — Research Intern

Mar – Sep 2025

Poladerme Deep Skin Reading, startup owned by ARCHOS group

Strasbourg, France

- Trained and benchmarked CNN backbones with transfer learning (EfficientNet-B6/B7, EfficientNetV2-L, ConvNeXt-L, ResNet-50) for **cancer lesion classification** on **4x NVIDIA A100 GPUs**, GPU-accelerated and parallelised, developed toward a **polarimetric imaging device**.
- Built a data pipeline over **74,000+ medical images** from public sources with metadata integration; preprocessing optimisation **cut runtime 30x** versus prior implementations.
- Ran a correlation study linking **polarimetric parameters to Mueller matrix coefficients**.

Computational Physics & Deep Learning on STM Imaging — Research Intern

Feb – Aug 2024

Institut Lumière Matière, UMR 5306 CNRS

Lyon, France

- Delivered an interactive tool identifying **borophene allotropes** from experimental STM images.
- Ran **DFT and molecular-dynamics** computation in VASP; automated deep-learning workflows over MD data in Python and Bash.

Opto-Electronic Detector Development & Photothermal Spectroscopy — Scientific Assistant (Intern)

May – Jun 2022

Institut Lumière Matière, UMR 5306 CNRS

Lyon, France

- Designed and built an opto-electronic detector** for photothermal deflection spectroscopy, quantifying probe-beam displacement.
- Implemented instrument interfacing and data acquisition in **LabVIEW**.

Electro-Optic System Test & Calibration — Optics R&D Intern

Jun – Jul 2021

Transvaro Electron Tools — defence electro-optics (thermal imagers, weapon sights)

Istanbul, Türkiye

- Calibrated opto-electronic systems** using a collimator and Infratest software.
- Set up a measurement environment for **Minimum Resolvable Temperature Difference (MRTD)**, **Non-Uniformity Correction (NUC)** and **boresight alignment** testing; organised inter-company meetings to resolve software issues.

EDUCATION

MSc Computer Science — Complementary Skills In Informatics

2025 – 2026

University Claude Bernard Lyon 1

Villeurbanne, France · GPA [2.5 / 4.00]

MSc Physics — Fundamental & Applied Physics, major in Optics & Photonics

2022 – 2024

University Claude Bernard Lyon 1

Villeurbanne, France · GPA [2.5 / 4.00]

Optics: lenses, filters, detectors, spectrometers, geometric / wave / Fresnel optics, optical instrumentation and spectroscopy. **Laser physics**: optical cavities, Gaussian beams, ultrafast and non-linear optics. **Experimental**: bench component calibration and alignment; microscopy (confocal, STED, SPIM, FRAP); AI for physics.

BSc — Science, Technology & Health, Physics track

2016 – 2021

University Claude Bernard Lyon 1

Villeurbanne, France · GPA [2.5 / 4.00]

TECHNICAL SKILLS

Optics & instrumentation

Optical component calibration and alignment · opto-electronic detector design and build-up · Mueller-matrix polarimetry · photothermal deflection spectroscopy · collimator setups · laser physics · spectrometers and detectors · LabVIEW instrument control and data acquisition

Imaging & machine learning

PyTorch, TensorFlow, Keras, scikit-learn, OpenCV · CNN transfer learning · semantic segmentation · multi-GPU training (NVIDIA A100) · Hugging Face · Fourier theory, frequency-domain filtering

Programming & computation

Python (NumPy, pandas, Matplotlib) · MATLAB / Simulink · C++ · Java · Bash · VASP (DFT, molecular dynamics) · PostgreSQL · pgadmin · Docker · Git · Linux / WSL · UML / Merise systems modelling (Modelio, Papyrus) · Arduino, Raspberry Pi, PID control · Wireshark

Extracurricular

Improv - clown · Fablab (electronics) Uni society member · Yoga · Cycling · Sailing · Scuba Diving (PADI)